



Smart Pen

Livescribe 3: A smart yet demanding smartpen for writers of all blends: <http://dgit.in/scribeaway>

Cloud? No cloud?

This article compares cloud computing and on-site computing pretty well with some good arguments for both: <http://dgit.in/cloudnocloud>

Industry Connect

Suit up in tech

Discussing wearable smart devices with Brian Bedrosian and Jeff Baer from Broadcom

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We had the chance to talk to two bigwigs at Broadcom and though they had no comments on their roadmap, they did tell us a bit about wearable smart devices (WSDs).

1 Which WSDs should we expect Broadcom to offer smart features to, apart from smart watches?

The wearable device market is still emerging and changing as new products are being introduced almost daily. While smart watches have been some of the earliest wearables to come to market, there is huge opportunity in the medical and fitness space for this type of technology. Some of the more interesting use cases include jewellery with proximity detection, baby socks that monitor sleep patterns, clothing that can cool consumers and more. These are just a few new areas wearable technology is headed, and Broadcom's technology is designed to support innovation in all categories.

2 Will the WSD solutions from Broadcom work only with Broadcom SOCs?

Broadcom technology is designed to be vendor-agnostic and work seamlessly with various solutions, including SOCs.

3 When do you think wearable technologies will no longer depend on a smartphone?

As wearable devices get smaller, lighter and more efficient, the connection to powerful smartphones or tablets that can process and send data to the cloud is critical. For this reason, smart mobile devices have become the hub for interconnecting everything. The value of a wearable device lies in its ability to connect to a smartphone or the Internet with minimal impact on battery life. Given Broadcom's

long-standing leadership in connectivity, we are uniquely positioned to widen business opportunities in this emerging space by offering the breadth of IP and customized components that enable creative new smart wearable devices to be connected.

4 Do you think we will see the integration of fitness devices such as the Nike Fuel Band and more with a smartwatch?

Fitness is an extremely important category for the growing Internet of Things and activity trackers are quickly gaining popularity in the market. Different from single-use or event-centric devices, activity trackers monitor multiple characteristics of the human body including movement, calories burned, body temperature, and sleep tracking, providing a new level of value to the consumer. Activity trackers are expected to grow at a 40 percent CAGR and overtake the 2013 shipment leader, heart rate monitors, in 2017.

5 Will we see a device from Broadcom competing with Qualcomm's 'Toq'? Can you talk about the details of the product?

We do not comment on competitor products or our future product roadmaps. What we can say is that given Broadcom's history of leadership in wireless connectivity and highly integrated offerings, we are well-positioned to capitalize on the market opportunity for wearables. Our advantage lies in integration, simplicity and cost.

6 What advantage do you think Wi-Fi Direct gives WSDs over NFC and Bluetooth?

The important thing here is to understand that different tools are meant for different purposes, and we support our customers no matter how they choose to add connectivity to wearable devices. Broadcom's expertise in Near Field Communication (NFC), Bluetooth Smart and Wi-Fi will help wearable devices to seamlessly connect to smart mobile devices and unlock new



Jeff Baer, Business Development Director, Embedded Wireless, Wireless Connectivity Combos

use cases. By using NFC, consumers can buy a new wearable device and simply tap it to their smartphone so the devices can instantly and securely establish a connection. Bluetooth Smart and Wi-Fi allow the consumer to collect data from their wearable device (i.e. calories burned, heart rate, etc.) and transfer it to their smartphone or into the cloud without draining the battery. By integrating Wi-Fi Direct into the WICED platform, we are enabling OEMs to rapidly develop wearable products that communicate seamlessly to the cloud using smartphones and tablets. By supporting Wi-Fi Direct, a specification that allows two devices to communicate with each other securely via Wi-Fi without an access point or computer, Broadcom is inspiring the development of new smart accessories, clothing and other wearable sensors.

7 In the face of criticism about the ambition and limited functionality of the smartwatch, what do you think is a perfect and practical WSD solution?

As the wearable market is in such an early stage and growing at such a rapid pace, the definition of a wearable is still being determined. However, what we see as critical success factors for the wearable market to take off include low-power, location-aware products that can connect anytime, anywhere. Wearables need to be small, comfortable, energy efficient and often fashionable. **1**